

Adithya Machavarapu

Data Analyst | Financial and Market Risk Analytics

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SUMMARY

Data Analyst specializing in financial and market risk analytics. Builds end-to-end analysis in Python, SQL and Excel, from raw data and cleaning through statistical testing and Value at Risk (VaR) modeling to dashboards and written reports, and backtests every model before its numbers are reported. HackerRank SQL (Advanced) certified; completed the Goldman Sachs Risk job simulation. B.Tech in Computer Science and Engineering (Data Analytics) at VIT-AP University, class of 2028.

TECHNICAL SKILLS

Programming and Databases: Python, SQL (MySQL): advanced joins, CTEs, window functions, query optimization

Data Analysis: pandas, NumPy, SciPy, exploratory data analysis (EDA), data cleaning and wrangling, statistical analysis, hypothesis testing, time-series analysis

Risk and Finance: Value at Risk (VaR: historical, parametric, Monte Carlo simulation), Expected Shortfall (ES), backtesting, stress testing, portfolio analytics

Visualization and Reporting: Microsoft Excel (advanced formulas, dashboards), Power BI, Matplotlib, Seaborn, KPI design, automated reporting

Tools: Git, GitHub, Jupyter Notebook, Streamlit, yfinance

PROJECTS

RiskScope: Market Risk Terminal

Python, Streamlit, yfinance | github.com/Adithyamachavarapu/Risk-Scope

- Built a Bloomberg-style market risk terminal with four modules (terminal, portfolio, options lab, reference): live market data, portfolio analytics, risk metrics and interactive financial charts.
- Computes Value at Risk and options analytics on streaming prices, so a position's risk updates the moment the market moves.

BlackRock Risk Lab: Equity Risk Report on Live Data

Python, Jupyter, pandas, Matplotlib | github.com/Adithyamachavarapu/blackrock-risk-lab

- End-to-end single-stock risk study on live data (2015 to present): return distributions, fat tails, Value at Risk, Expected Shortfall and valuation, with eight publication-ready charts.
- Every figure is reproducible from the notebook, so the analysis can be audited line by line.

Credit Risk Analysis: 32,000 Loans in Pure SQL

SQL, MySQL | github.com/Adithyamachavarapu/credit-risk-analysis

- Analyzed 32,000 real loans entirely in SQL: default rates by borrower segment, risk-based pricing checks and cohort views, using CTEs and window functions throughout.
- Findings sit beside the exact queries that produced them, so every number is verifiable.

UPI Transaction Risk: Statistical Investigation

Python, pandas, SciPy | github.com/Adithyamachavarapu/upi-transaction-risk

- Analyzed 1,000 UPI payment records to explain a high failure rate; applied significance testing to separate genuine risk patterns from random noise.
- Concluded, and reported plainly, that the failures were statistically indistinguishable from noise, choosing an honest finding over a dramatic one.

EDUCATION

VIT-AP University, Amaravati, B.Tech, Computer Science and Engineering (Data Analytics)

Expected 2028

CERTIFICATIONS

SQL (Advanced), HackerRank Skills Certification

Sep 2026

Certificate ID 2639C498C94C

Risk Job Simulation, Goldman Sachs (via Forage)

Aug 2026

Introduction to risk; evaluating client profiles and real-estate investments.